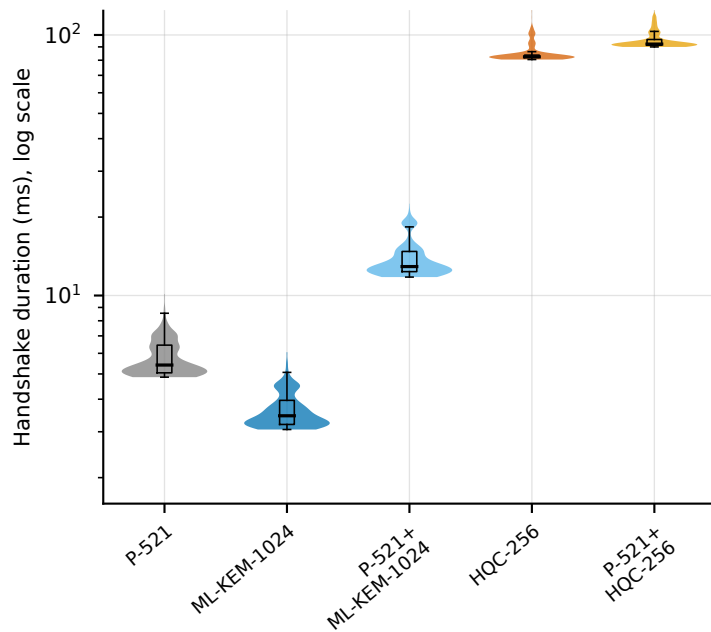
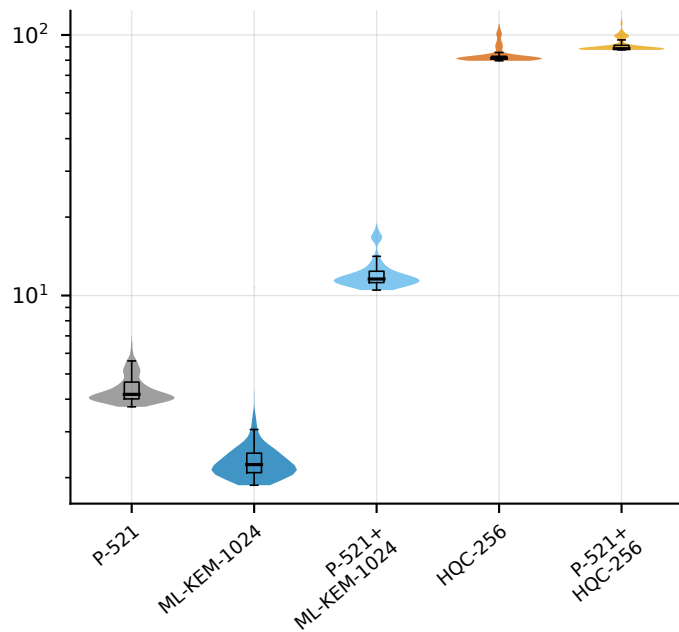


(a)

secp521r1 (classical)

**(b)**

mlds87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mlds87 Mean (ms)	mlds87 CV (%)	mlds87 Max (ms)
P-521	5.84	16.2	10.1	4.42	13.1	7.00
ML-KEM-1024	3.67	15.8	6.06	2.34	21.3	10.8
P-521+ML-KEM-1024	13.9	16.3	22.5	12.3	14.6	18.9
HQC-256	85.5	8.4	128.7	84.3	7.7	110.4
P-521+HQC-256	95.6	7.7	130.6	91.4	6.1	122.8